

Evidence review

The harness (still) matters

Public studies show that the agent harness changes performance, cost, token use and runtime. They do not identify one harness that wins across every model and task.

Evidence captured on 7 August 2026

Current observations

- **Harness and runtime choices can materially change results.** Claw-SWE-Bench found 12.5- to 27.4-point spreads; Harvey's intervention ranged from 0.9 points worse to 23 points better.
- **No harness wins across all models and tasks.** Native CLIs led 4 of 6 current Terminal-Bench pairs, while alternative harnesses led elsewhere.
- **Efficiency differences are often clearer than quality differences.** Databricks, GitHub and several smaller studies found large cost, context or token gaps beside modest quality changes.
- **The count is not 24 independent replications.** Some studies reuse benchmark families or live leaderboards, and many rely on a single attempt per task.
- **We separate matched harness evidence from same-label system comparisons.** Provider routing and model snapshots can still differ even when the published model label matches.
- **Teams should test harnesses on their own work.** Compare at least 2 harnesses with the same model, effort and budget. Measure quality, cost and time.

Currently, we know of 24 public harness comparisons and diagnostics

23 hold the model fixed and measure task quality. One isolates prompt overhead.

1

Operational and broad evidence

Start here: larger, more operational or more extensively reported comparisons.

01	Databricks coding-agent benchmark	8 Jul
02	GitHub Copilot harness evaluation	25 Jun
03	Terminal-Bench 2.1	6 May
04	Coding Agent Index	11 May
05	Agents' Last Exam	3 Jun
06	DrugDiscoveryBench	29 Jun
07	Claw-SWE-Bench	10 Jun
08	Harness-Bench paper	27 May
09	WANDR system comparison and delivery ablation	14 Jul
10	Warden benchmark	5 Jun
11	Agent Security League	9 Jul

2

Focused domain evidence

Strong task-specific evidence, but conclusions may not transfer outside the domain.

12	Harvey LAB harness ablation	22 Jun
13	Harness optimization on LAB	1 Jul
14	LatchBio behavioural-prior study	10 Jul
15	CSI scaffold comparison	27 May
16	Vulnerability-Agent-Bench	1 Feb
17	WildClawBench	11 May
18	CORE-Bench Hard	23 Jun

3

Small and diagnostic evidence

Useful signals with limited task counts or no direct measure of coding quality.

19	The Scaffold Effect	29 Apr
20	HAL SWE-bench Verified Mini	13 Oct 2025
21	HarnessBench	8 May
22	A-CODE-CLI	29 Jun
23	OpenBench M4.5	3 Jul
24	Harness Tax	13 Apr

What each benchmark tells us

1

Operational and broad evidence

01	Databricks coding-agent benchmark Across 6 matched comparisons, Claude Code or Codex cost 1.20 to 2.08 times as much as Pi. Quality stayed within 7 points.
02	GitHub Copilot harness evaluation Across 20 matched pairs, Copilot CLI moved task resolution by -8.0 to +7.1 points and used fewer median tokens in 18.
03	Terminal-Bench 2.1 The native CLI led 4 of 6 current matched pairs; 1 was effectively tied and Terminus 2 led Gemini 3 Pro.
04	Coding Agent Index OpenCode led with Opus 4.7; Codex led Cursor with GPT-5.5. In both comparisons, the higher-scoring harness cost more.
05	Agents' Last Exam GPT-5.5 full-pass rates spanned 19.1% to 24.0%, while published total cost spanned \$180 to \$570.
06	DrugDiscoveryBench mini-SWE-agent scored highest with GPT-5.5. Codex placed in the middle.
07	Claw-SWE-Bench The harness spread was 12.5 points with GLM-5.1 and 27.4 points with Qwen3.6-Flash.
08	Harness-Bench paper Across the same 8-model pool, average combined score ranged from 52.4 for OpenClaw to 76.2 for NanoBot.
09	WANDR system comparison and delivery ablation A contextual production-system comparison showed a large quality-cost trade-off; within-system ablations favoured file or share delivery over output tokens.
10	Warden benchmark Pi found a similar or larger share of known vulnerabilities at about one-quarter to one-fifth of Claude SDK's cost.
11	Agent Security League Cursor led on Opus 4.7 and 4.8. Claude Code led on Sonnet 5.

2

Focused domain evidence

12	Harvey LAB harness ablation The same harness intervention ranged from 0.9 points worse to 23 points better, depending on the model.
13	Harness optimization on LAB With DeepSeek V4 Pro fixed, pooled criterion pass rate ranged from 3.5% to 80.1% across 5 harnesses.
14	LatchBio behavioural-prior study Point-estimate ordering changed by task horizon, although the harness intervals overlapped; a prompt-only ablation moved in opposite directions.
15	CSI scaffold comparison CSI::Claude and CSI::Codex both solved 15 of 33 tasks, but Codex cost one-third as much and finished faster.
16	Vulnerability-Agent-Bench Claude Code and Codex beat OpenCode on all 3 published matched-model pairs.
17	WildClawBench The best harness changed by model. Switching harness moved scores by double digits.
18	CORE-Bench Hard Claude Code more than doubled CORE-Agent's verified success with Opus 4.5 while costing less.

3

Small and diagnostic evidence

19	The Scaffold Effect Goose used about 41 times fewer reported tokens than OpenCode for similar pass rates on 2 fixed models.
20	HAL SWE-bench Verified Mini SWE-Agent usually solved more tasks than HAL Generalist, but often at several times the reported cost.
21	HarnessBench Several configurations finished within one task of the leader. Runtime differed more clearly.
22	A-CODE-CLI OpenCode had a small backend lead over Claude Code and cost 44% less with Sonnet 4.6.
23	OpenBench M4.5 All 4 harnesses passed every task. Pi and Cursor used less time and fewer tokens.
24	Harness Tax Fixed input overhead ranged from about 2,600 tokens for Pi to 27,000 for Claude Code.

Databricks ran 6 matched harness comparisons: Claude Code or Codex cost 1.20 to 2.08 times as much as Pi, with quality within 7 points.

What they did

Databricks replayed recent human-written pull requests from its multi-million-line internal codebase. It compared Claude Code with Pi on Opus 4.8 and Codex with Pi on GPT-5.5 at matched effort settings.

Tasks	Not published
Repeated trials	not clear
Grader	Repository tests reviewed by engineers

What we observe

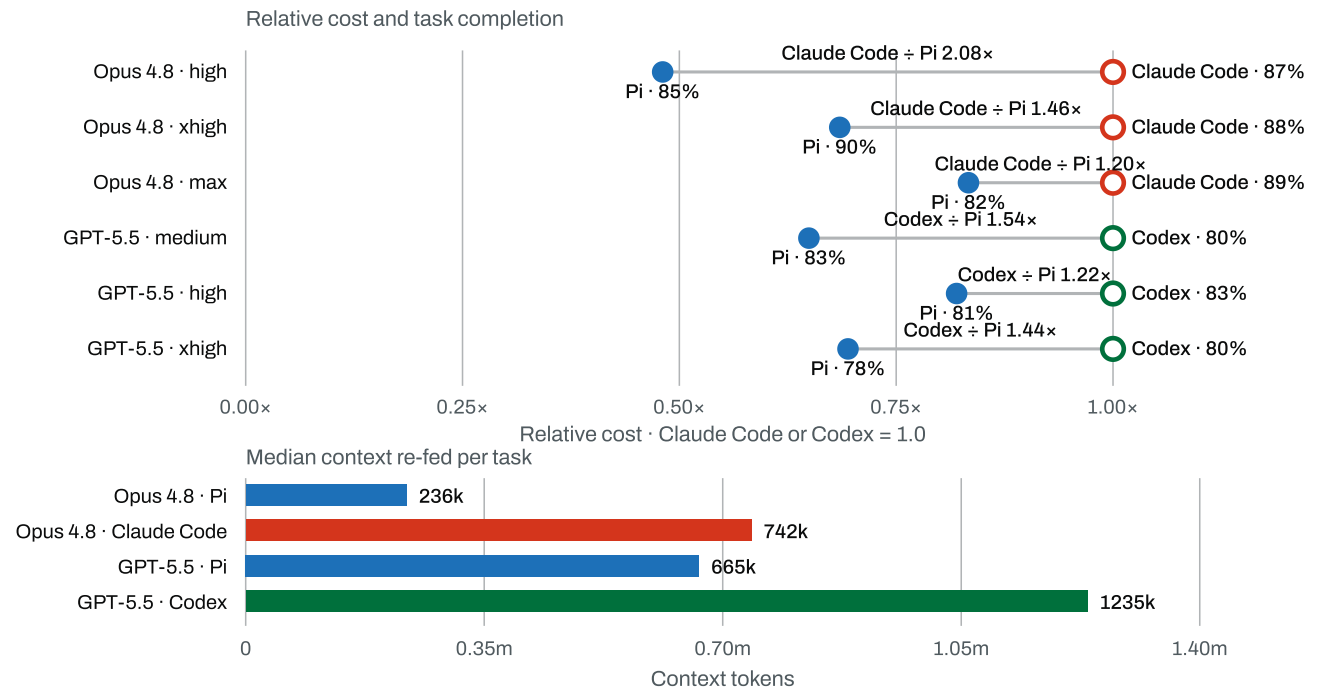
Pi was cheaper in all 6 published pairs. It also re-fed less context: 236,000 versus 742,000 tokens on Opus 4.8 and 665,000 versus 1.235 million on GPT-5.5.

Limit

The source does not publish task count, run-level data or uncertainty. Per-task dollar costs: three combinations are stated exactly in the text; the rest are derived from the published cost ratios or traced from Figure 1.

Databricks results

Published results



The source publishes exact cost ratios and scores, but not exact dollar values for each matched point. Costs are indexed to Claude Code for Opus and Codex for GPT.

GitHub compared Copilot CLI with Claude Code or Codex across 5 suites: quality was mixed, while Copilot used fewer median tokens in 18 of 20 pairs.

What they did

GitHub fixed the model, task, context limit, reasoning effort and environment across Copilot CLI and each model provider's native harness. It tested 4 models on 5 suites and ran each configuration 5 times.

Tasks	Not published
Repeated trials	yes
Grader	Public benchmark graders and internal executable checks

What we observe

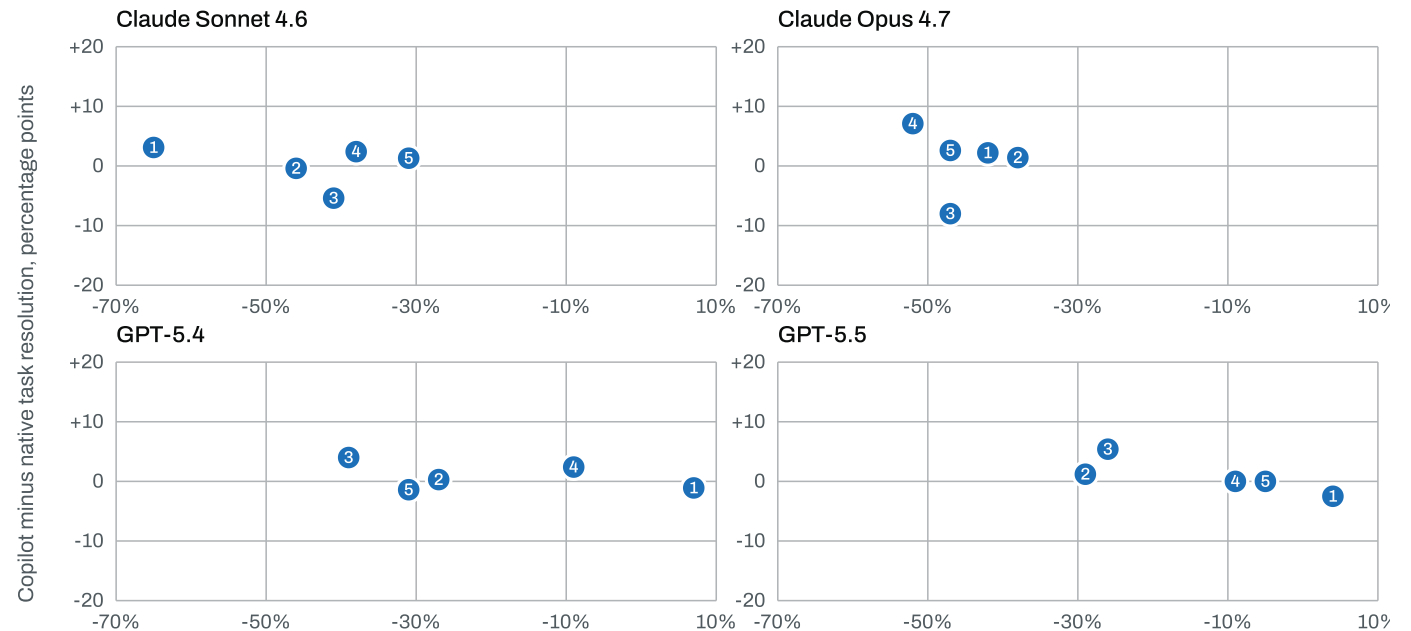
Task-resolution differences ranged from 8.0 points worse to 7.1 points better. Copilot used fewer median tokens in 18 of 20 model and benchmark pairs.

Limit

This is a vendor study with no raw result rows. For suites under 100 tasks, GitHub reports the best of 5 runs while labelling the metric pass@1.

GitHub results

Published results



Copilot median token change · negative means fewer tokens

1 SWE-bench Verified · 2 SWE-bench Pro · 3 SkillsBench · 4 Win-Hill · 5 Terminal-Bench 2. Axes include zero; the task-resolution axis spans ± 20 points, wider than the observed range. GitHub says the quality differences are within run-to-run variance.

Terminal-Bench repeated 89 command-line tasks: the native CLI led 4 of 6 current matched pairs, but not for every model.

What they did

Terminal-Bench runs 89 containerized terminal tasks with deterministic final-state tests. The current 2.1 leaderboard contains 6 clean same-model pairs between a provider CLI and Terminus 2, with at least 5 runs per configuration.

Tasks	89
Repeated trials	yes
Grader	Deterministic final-state tests

What we observe

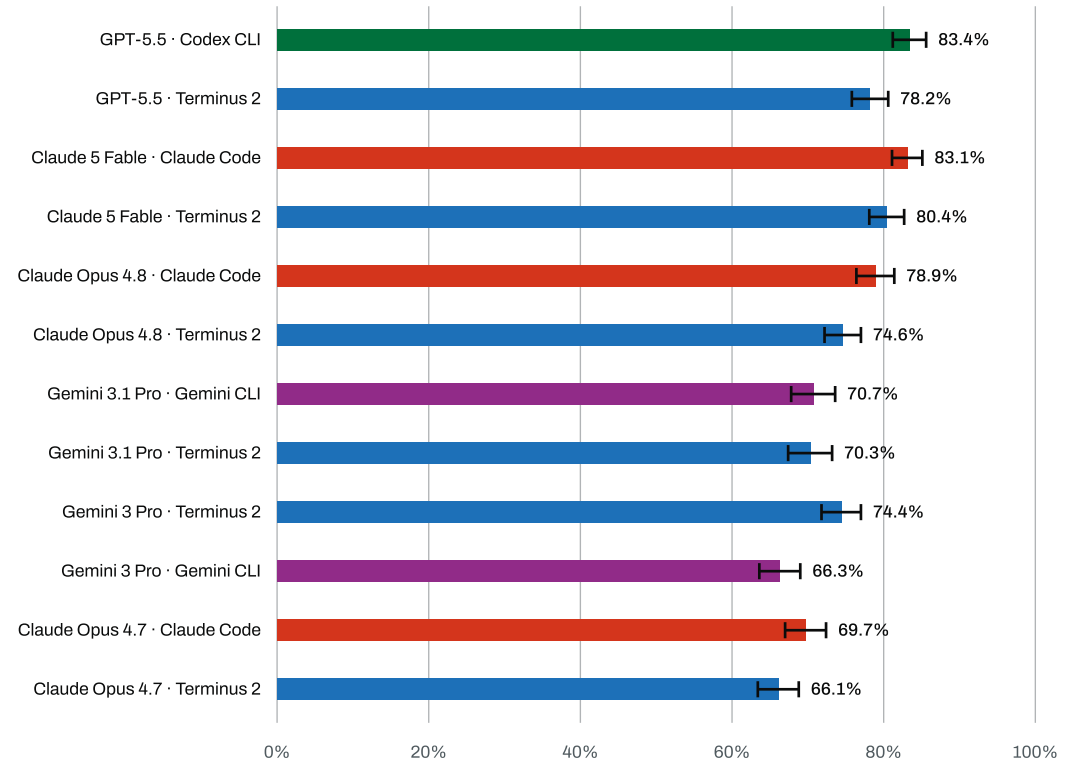
Claude Code or Codex led 4 pairs. Gemini 3.1 Pro was effectively tied, while Terminus 2 led Gemini 3 Pro by 8.1 points.

Limit

The leaderboard changes over time. A token appendix in the 2.0 paper says it covers 74 tasks, not the full 89, so this review does not derive per-task token values from it.

Laude Institute results
Dataset or repository

Published results



Whiskers reproduce the leaderboard's published 95% confidence intervals. The scale runs from 0 to 100%.

Artificial Analysis ran 321 tasks 3 times: the higher-scoring harness depended on the model and cost more in both full-suite comparisons.

What they did

Artificial Analysis combines 321 tasks from DeepSWE, Terminal-Bench v2 and SWE-Atlas-QnA. It repeats each configuration 3 times and reports an equally weighted index, API cost and active wall time.

Tasks	321
Repeated trials	yes
Grader	Program checks, test suites and rubric scoring

What we observe

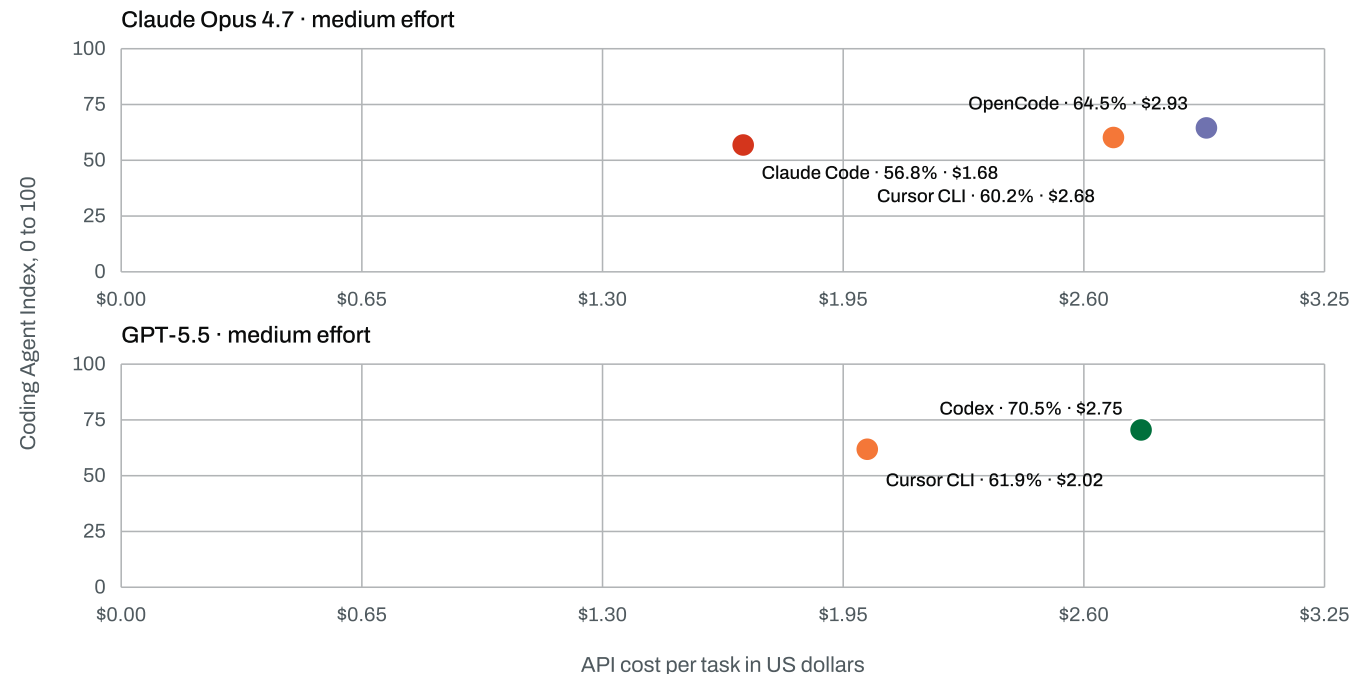
OpenCode led the Opus 4.7 group by 7.7 points; Codex led Cursor with GPT-5.5 by 8.7 points. Most of both gaps came from DeepSWE.

Limit

No uncertainty or task-level results are published. The live suite, pricing and harness versions change. The GPT-5.5 match relies on the published model and effort labels.

Artificial Analysis results

Published results



The live page also has a GPT-5.4 pair with only 2 of the current 3 component benchmarks. It is in the downloadable observations but excluded from these full-suite panels.

Agents' Last Exam tested long professional workflows: GPT-5.5 harnesses stayed within 4.9 points, but total cost differed by more than 3 times.

What they did

Agents' Last Exam evaluates long professional workflows with runs capped at 5 hours. It reports fixed-model harness sweeps for GPT-5.5 and Opus 4.7, plus a 105-task ALE-CLI sweep for Sonnet 4.6.

Tasks	152
Repeated trials	not clear
Grader	Task-specific executable and rubric-based checks

What we observe

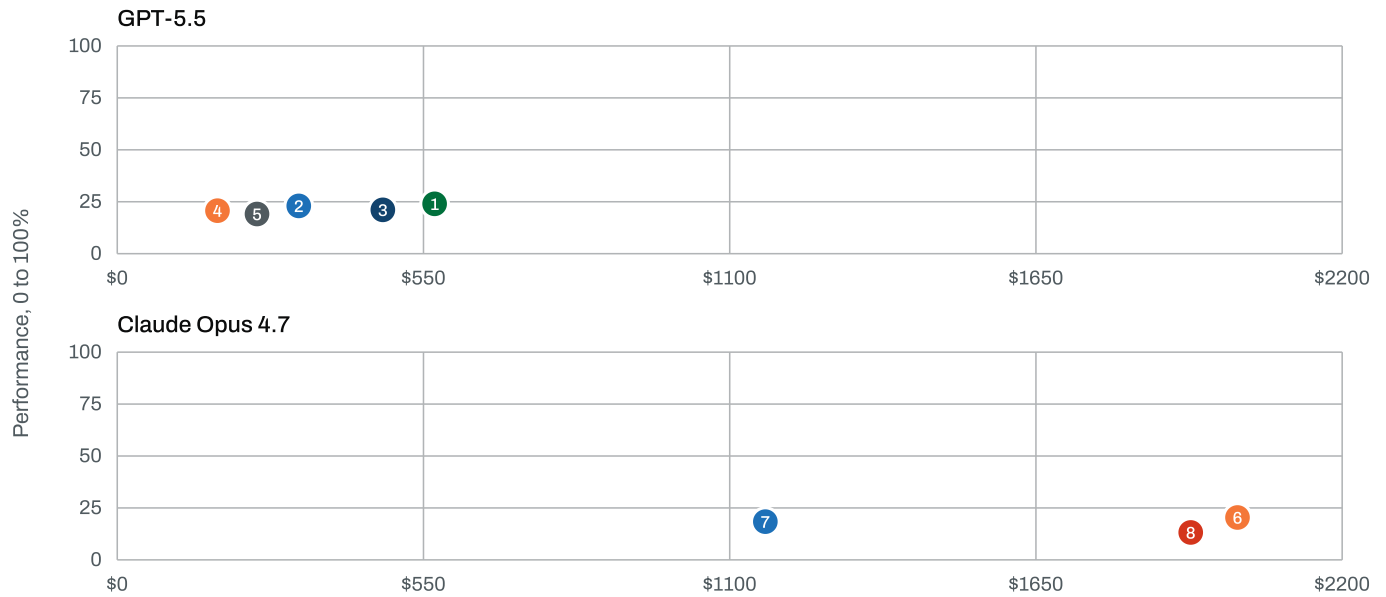
The harness changed both full-pass rate and resources, but the model effect could be larger: GPT-5.5 harnesses were within 4.9 points while the fixed-OpenClaw model spread was 16.8 points.

Limit

Only some configurations have 3 independent runs. The published tier counts do not add cleanly to the stated 152-task public set.

RDI Berkeley results
Dataset or repository

Published results



Published total evaluation cost in US dollars				Published total evaluation cost in US dollars			
Point	Model	Harness	Result · cost	Point	Model	Harness	Result · cost
1	GPT-5.5	Codex	24% · \$570.00	5	GPT-5.5	Droid	19.1% · \$251.00
2	GPT-5.5	ALE-Claw	23% · \$326.00	6	Claude Opus 4.7	Cursor	20.4% · \$2012.00
3	GPT-5.5	OpenClaw	21.1% · \$477.00	7	Claude Opus 4.7	ALE-Claw	18.4% · \$1164.00
4	GPT-5.5	Cursor	20.7% · \$180.00	8	Claude Opus 4.7	Claude Code	13.2% · \$1928.00

The panels use the source's stated 152-task public set. Sonnet 4.6 ALE-CLI rows use a different 105-task subset and remain in the downloadable data.

DrugDiscoveryBench repeated 82 expert workflows: mini-SWE-agent led the matched GPT-5.5 comparison, while Codex placed in the middle.

What they did

Scale AI ran 82 expert-curated drug-discovery and life-sciences workflows 3 times. An LLM judge scored each result against expert-authored rubrics and reference answers.

Tasks	82
Repeated trials	yes
Grader	Expert-authored rubrics scored by an LLM judge

What we observe

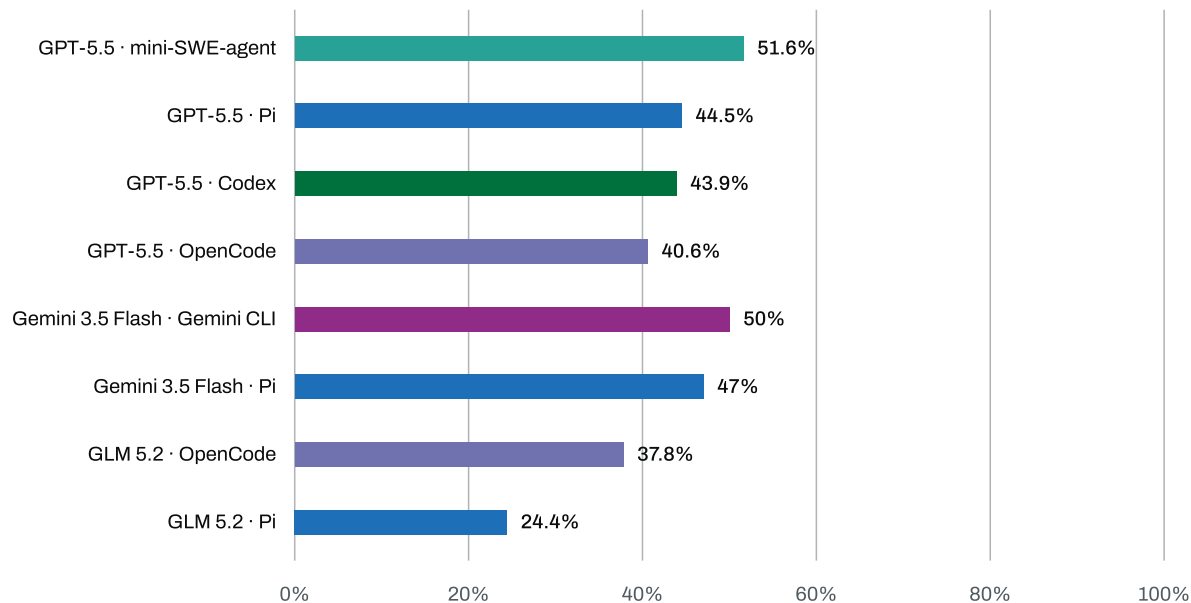
GPT-5.5 performed best in mini-SWE-agent. Codex was neither the best nor worst harness for that model.

Limit

The paper plots cost but does not publish a table of exact cost points. The outcome depends on an LLM judge.

Scale AI results
Dataset or repository

Published results



Claw-SWE-Bench crossed 5 harnesses with 2 models on 350 issues: the harness spread grew from 12.5 to 27.4 points depending on the model.

What they did

Claw-SWE-Bench fully crossed 5 harnesses and 2 models on the same 350 multilingual GitHub issues, with a fixed prompt, workspace contract, runtime budget, patch extraction and evaluator.

Tasks	350
Repeated trials	no
Grader	Executable issue-resolution tests

What we observe

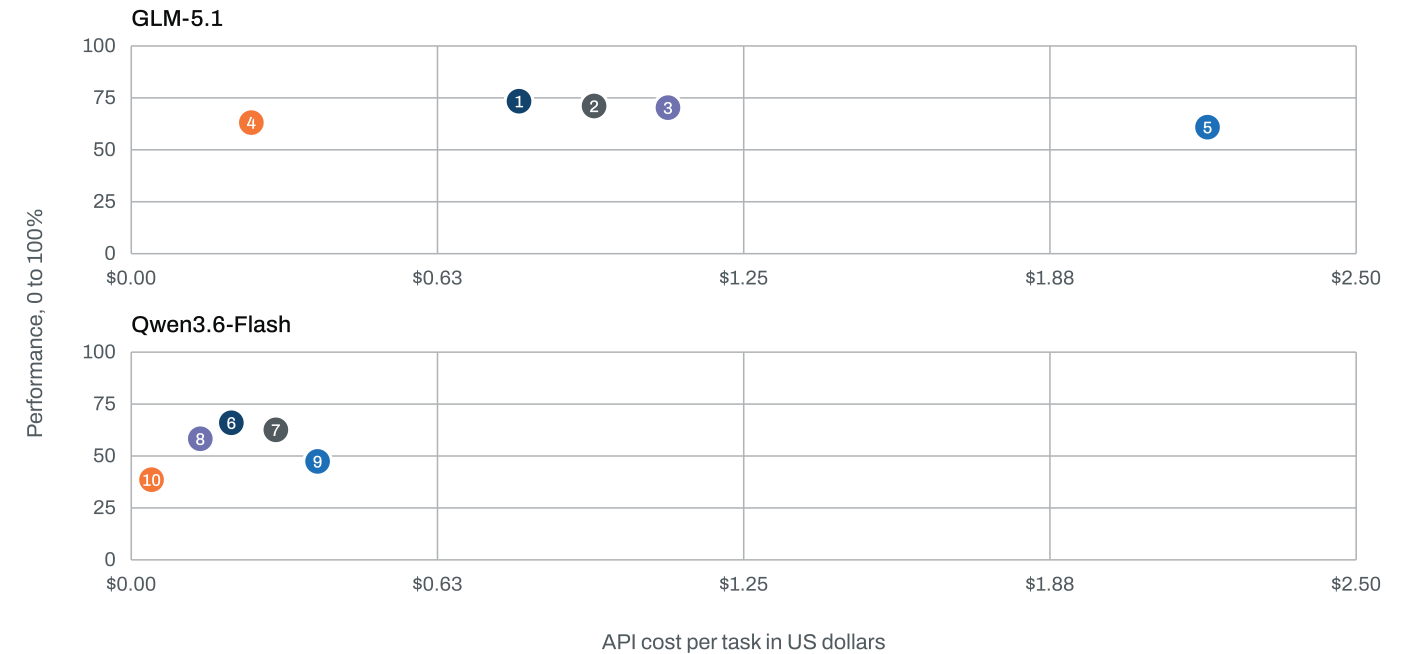
OpenClaw led both model groups. The gap between best and worst harness was 12.5 points for GLM-5.1 and 27.4 points for Qwen3.6-Flash.

Limit

There is one attempt per task and configuration, with no confidence intervals. Token accounting contains large harness-specific anomalies.

OpenSquilla results
Dataset or repository

Published results



Point	Model	Harness	Result · cost	Point	Model	Harness	Result · cost
1	GLM-5.1	OpenClaw	73.4% · \$0.79	6	Qwen3.6-Flash	OpenClaw	66% · \$0.20
2	GLM-5.1	Hermes	71.1% · \$0.94	7	Qwen3.6-Flash	Hermes	62.6% · \$0.30
3	GLM-5.1	ZeroClaw	70.3% · \$1.10	8	Qwen3.6-Flash	ZeroClaw	58.3% · \$0.14
4	GLM-5.1	GenericAgent	63.1% · \$0.25	9	Qwen3.6-Flash	NanoBot	47.4% · \$0.38
5	GLM-5.1	NanoBot	60.9% · \$2.20	10	Qwen3.6-Flash	GenericAgent	38.6% · \$0.04

Cost per task is the published configuration total divided by the common 350-task set. Every panel and axis starts at zero.

Harness-Bench ran a 6-harness by 8-model factorial: average combined score ranged from 52.4 to 76.2 across harnesses.

What they did

Harness-Bench ran 6 configurable harnesses with 8 models on 106 executable professional tasks, producing 5,088 trajectories. Only 22 tasks are software engineering.

Tasks	106
Repeated trials	no
Grader	Executable checks plus process judging

What we observe

NanoBot had the highest harness-average combined score and completion rate. The size and direction of the harness effect still changed by model.

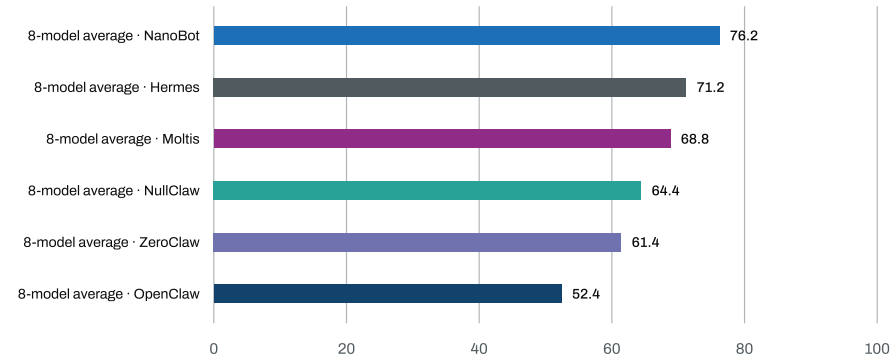
Limit

There is one trajectory per task and configuration. Combined score includes process judging, and the public repository currently has no licence file.

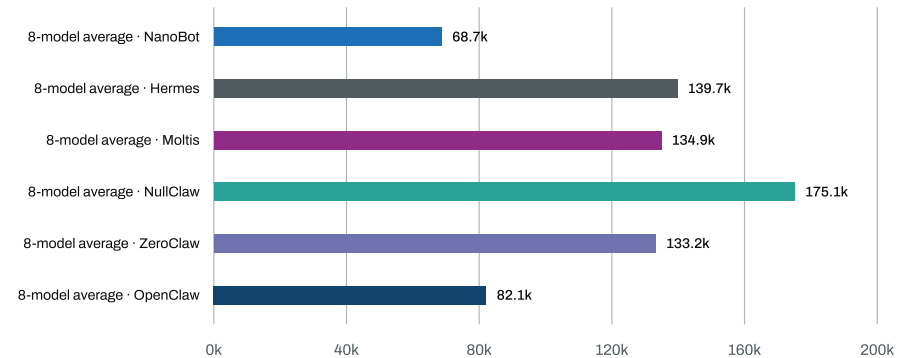
Qihoo 360 results
Dataset or repository

Published results

Combined score



Tokens per task



Each bar averages the same 8-model pool. Combined score includes process judging as well as task completion.

WANDR compared 500 research runs across 2 production systems, then isolated output delivery on 45 tasks: file or share delivery scored higher in all 3 systems.

What they did

Perplexity compared Search as Code and OpenAI Responses, both labelled GPT-5.5, across 500 tasks and a 45-task effort sweep. Separately, it held each provider route and effort fixed while changing only whether results were delivered through files or output tokens.

Tasks	500
Repeated trials	no
Grader	WANDR record-level verifier with pinned GPT-5.4 judges

What we observe

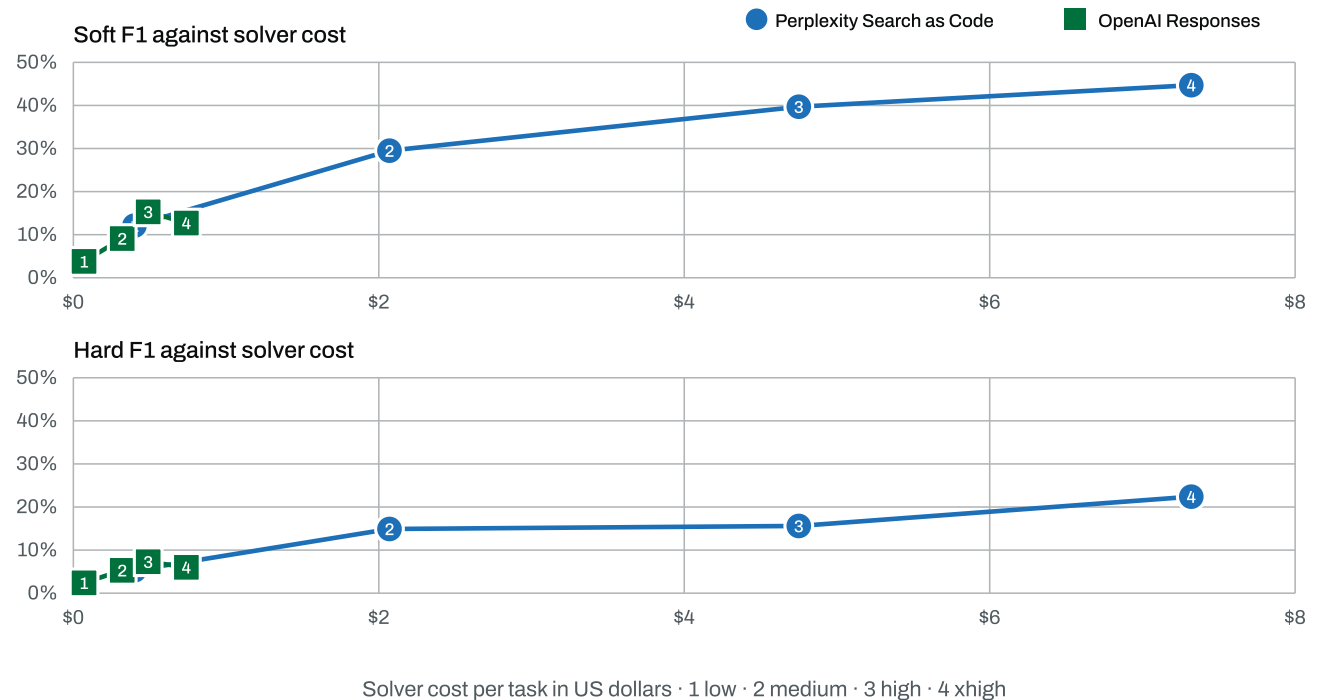
The production systems showed a large quality-cost trade-off but do not isolate a harness effect. Within each of Perplexity, Anthropic and OpenAI, file or share delivery scored above output-token delivery.

Limit

The production systems may use different model snapshots, search infrastructure and tools. The paper reports one rollout per task, no confidence intervals and no evaluated run artefacts; delivery-ablation arms are independent rather than paired rollouts.

Perplexity results
Dataset or repository

Published results



Both axes start at zero. In the separate 500-task high-effort run, Search as Code scored 36.3% soft F1 at \$5.20 per task; OpenAI Responses scored 12.1% at \$0.50. A delivery ablation also favoured file or share delivery in all 3 systems, but used independent rollouts.

Sentry tested 86 known vulnerabilities: Pi found a similar or larger share while costing about one-quarter to one-fifth as much as Claude SDK.

What they did

Sentry gave agents files connected to 86 historical vulnerabilities and measured how many known findings they recovered. The published table compares matched Claude models running in Pi and Claude SDK.

Tasks	86
Repeated trials	no
Grader	Known-corpus recall

What we observe

Pi cost less and found a similar or larger share of the known corpus for the matched Claude models.

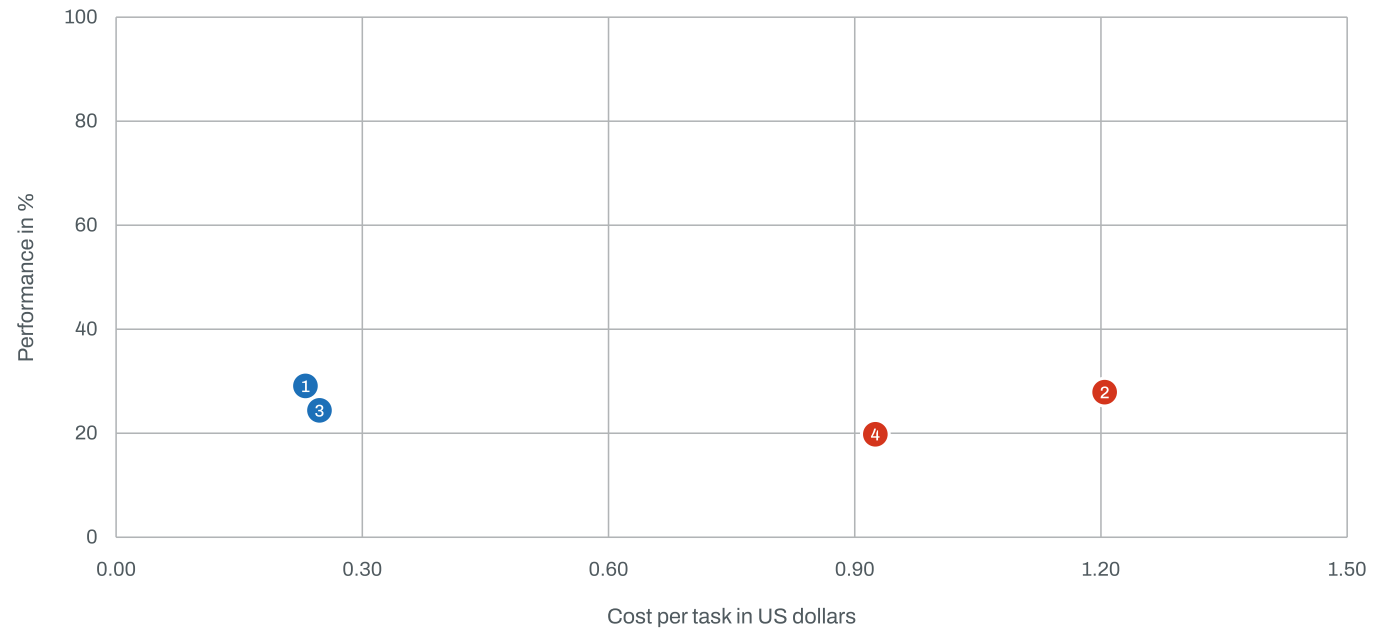
Limit

The comparator is Claude SDK, not the Claude Code command-line product. The benchmark does not measure whole-repository discovery.

Sentry results

Dataset or repository

Published results



Point	Model	Harness	Result · cost
1	Claude Sonnet 4.6	Pi	29.1% · \$0.23
2	Claude Sonnet 4.6	Claude SDK	27.9% · \$1.20
3	Claude Opus 4.8	Pi	24.4% · \$0.25
4	Claude Opus 4.8	Claude SDK	19.8% · \$0.93

Endor Labs tested 200 vulnerability repairs: Cursor led for Opus, but Claude Code led for Sonnet 5.

What they did

Endor Labs asked each configuration to repair 200 real vulnerabilities. Hidden tests separately measured whether the software still worked and whether the vulnerability was fixed.

Tasks	200
Repeated trials	no
Grader	Hidden functional and security tests

What we observe

Cursor led Claude Code for Opus 4.7 and 4.8. Claude Code led Cursor for Sonnet 5.

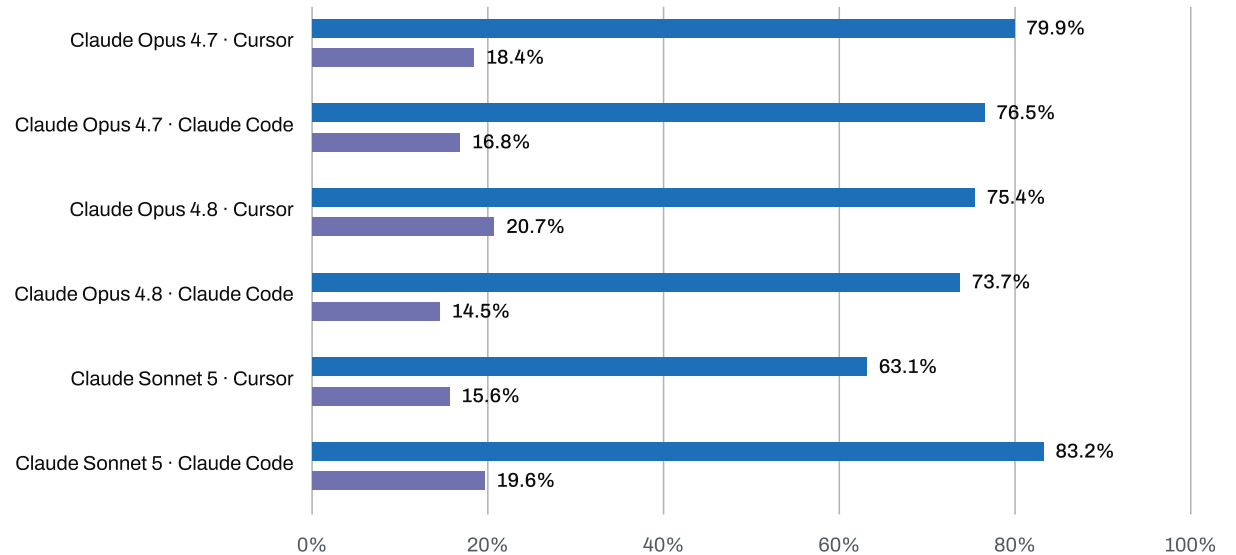
Limit

There are no repeated trials or task-level costs. Timeouts explain part of the reversal.

Endor Labs results

Published results

Functional pass rate Security pass rate



Harvey tested 180 legal tasks: the Applied Compute harness ranged from 0.9 points worse to 23 points better across 6 fixed models.

What they did

Harvey compared its baseline LAB-style harness with an Applied Compute harness on 180 held-out legal tasks and about 10,000 criteria. It held 6 models and the grader fixed.

Tasks	180
Repeated trials	no
Grader	GPT-5 Mini against expert criteria, graded 3 times and averaged

What we observe

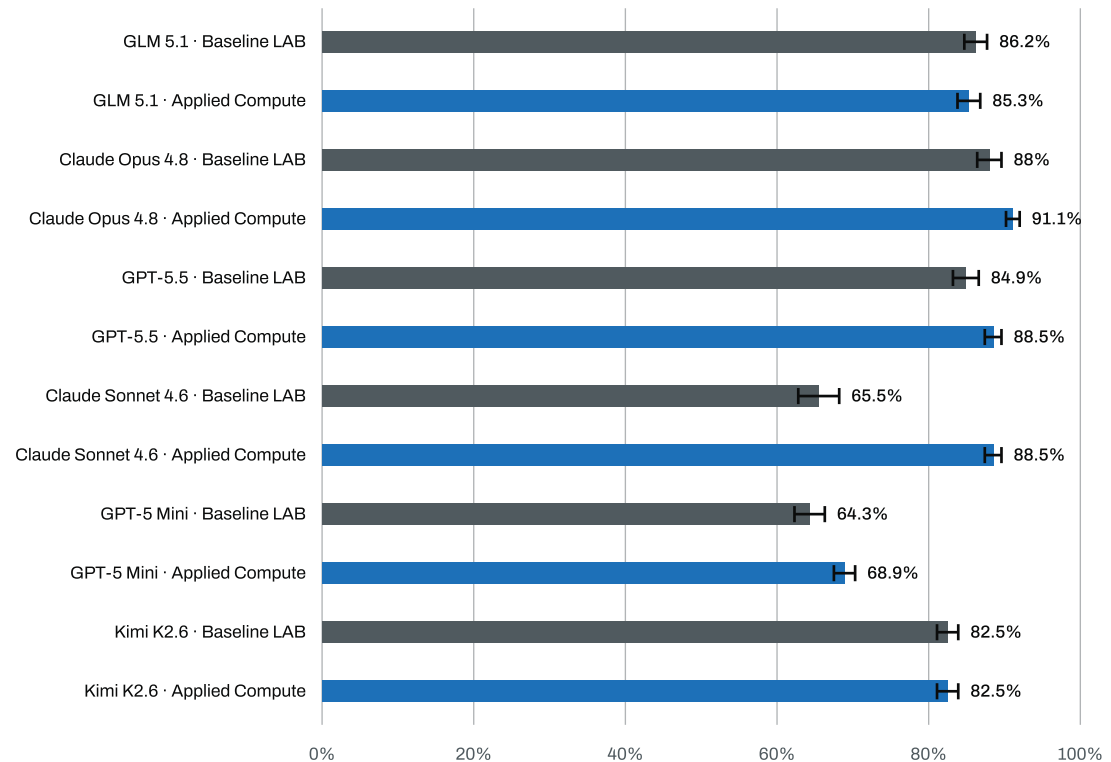
The treatment effect was heterogeneous: slightly negative for GLM 5.1, neutral for Kimi K2.6 and strongly positive for Sonnet 4.6.

Limit

The exact test split and Applied Compute harness are not public. Published error values are not defined, and agent cost, runtime and tokens are not reported.

Harvey and Applied Compute results
Dataset or repository

Published results



Whiskers reproduce the source's published error values. Harvey does not define them as confidence intervals. The scale runs from 0 to 100%.

LAB held DeepSeek V4 Pro fixed on 100 legal tasks: changing the harness moved pooled criterion pass rate from 3.5% to 80.1%.

What they did

The study froze DeepSeek V4 Pro, a 100-task held-out LAB split and the Sonnet 4.6 judge. It compared an automatically optimized LAB harness, vanilla LAB, Pi, Goose and mini-swe-agent.

Tasks	100
Repeated trials	no
Grader	LAB criterion rubrics scored by Claude Sonnet 4.6

What we observe

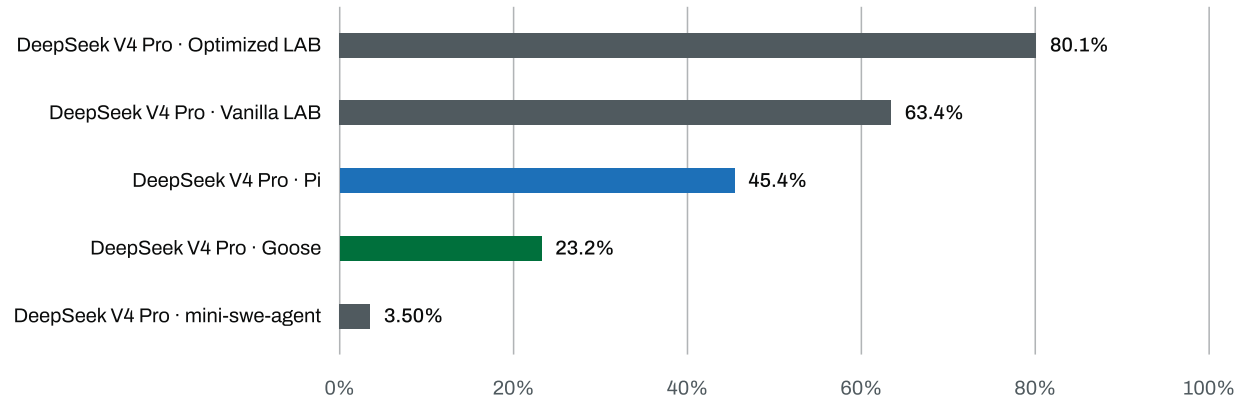
Harness choice dominated the measured result. Deterministic file-delivery and runtime fixes produced the largest gains, while general coding harnesses struggled with LAB's deliverable contract.

Limit

The judge is model-based, test configurations were run once, and the optimized harness was selected after repeated development on a separate 24-task split. Timeouts and output-delivery failures remained scored as harness failures.

Joel Niklaus, Hugging Face results
Dataset or repository

Published results



LatchBio repeated 121 biology tasks: Pi's point estimate was 4.6 points higher on shorter work, while Claude Code's was 4.8 points higher on long workflows; intervals overlapped.

What they did

LatchBio ran 100 TxBench-PP tasks and 21 scBench-Long workflows 3 times per configuration. It compared Pi with native harnesses, then held Opus 4.8, Pi's tools, tasks and graders fixed while adding calibration and orchestration instructions to Pi's system prompt.

Tasks	121
Repeated trials	yes
Grader	Deterministic biological graders

What we observe

The point-estimate ordering changed with task horizon, but published harness intervals overlapped on both suites. In the prompt-only ablation, orchestration guidance reduced TxBench-PP pass rate by 4.7 points and increased scBench-Long pass rate by 3.2 points.

Limit

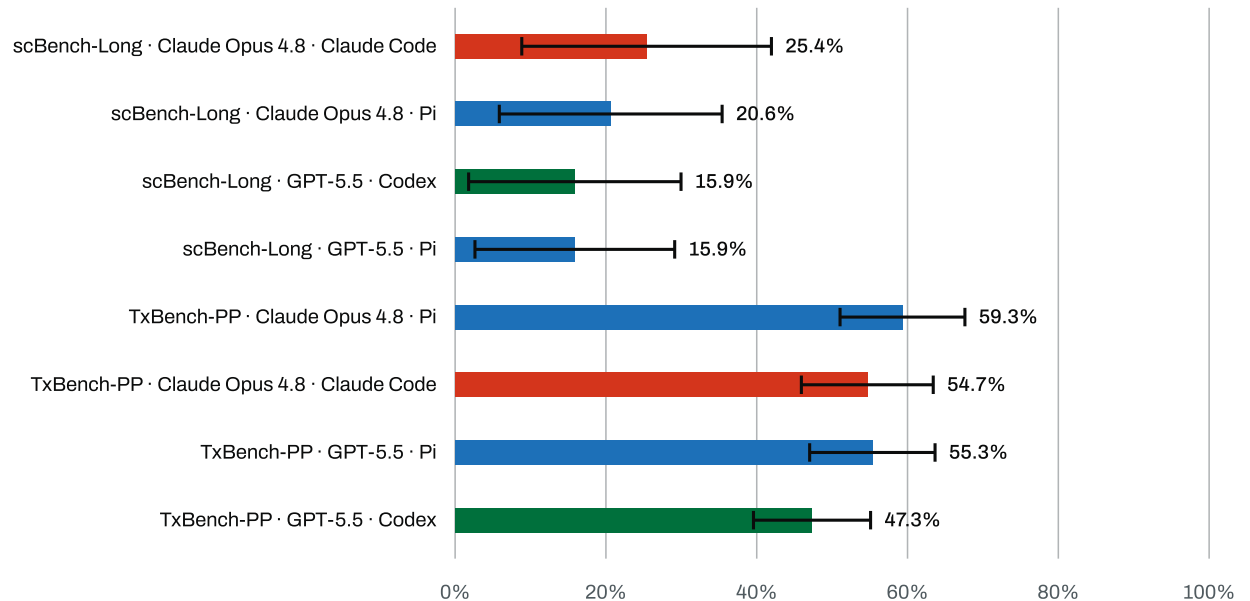
The harness comparisons do not establish a winner because their intervals overlap, and the 21-task long-horizon intervals are especially wide. Prompt-ablation results are aggregate-only, and the behavioural clauses approximate rather than reproduce Claude Code's full prompt.

LatchBio results

Dataset 1

Dataset 2

Published results



CSI ran 5 scaffolds with one model on 33 security challenges: Claude and Codex tied on solves, while Codex cost one-third as much.

What they did

Alias Robotics ran 5 scaffolds with the same private alias2-mini model on 33 CyBench capture-the-flag challenges and recorded solves, total API cost, wall time, commands and tokens.

Tasks 33

Repeated trials no

Grader Challenge flag verification

What we observe

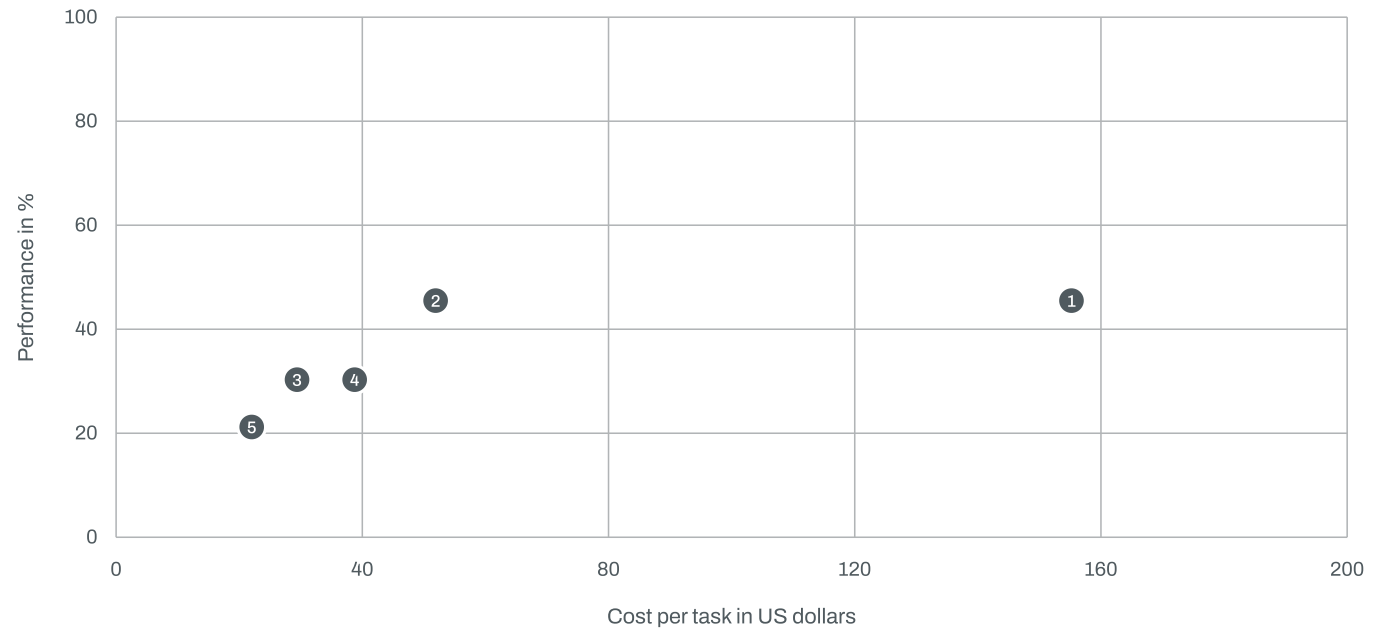
CSI::Claude and CSI::Codex both solved 15 tasks. CSI::Codex used one-third of the published cost and 31% less wall time.

Limit

There is one run per scaffold and challenge. The model is proprietary, the task set is small and no dedicated experiment artefact is public.

Alias Robotics results

Published results



Point	Model	Harness	Result · cost
1	alias2-mini	CSI::Claude	45.5% · \$155.21
2	alias2-mini	CSI::Codex	45.5% · \$51.91
3	alias2-mini	CSI::Mistral	30.3% · \$29.39
4	alias2-mini	CSI::GCAI	30.3% · \$38.76
5	alias2-mini	CSI::CAI	21.2% · \$22.03

XOR tested 136 vulnerability repairs: Claude Code and Codex beat OpenCode on all 3 published matched-model pairs.

What they did

XOR ran agents in isolated containers on real vulnerability-repair tasks. Its public explorer reports pass rate and cost for matched model and harness configurations.

Tasks	136
Repeated trials	no
Grader	Vulnerability repair checks

What we observe

Claude Code and Codex led the corresponding OpenCode configurations on the published matched pairs.

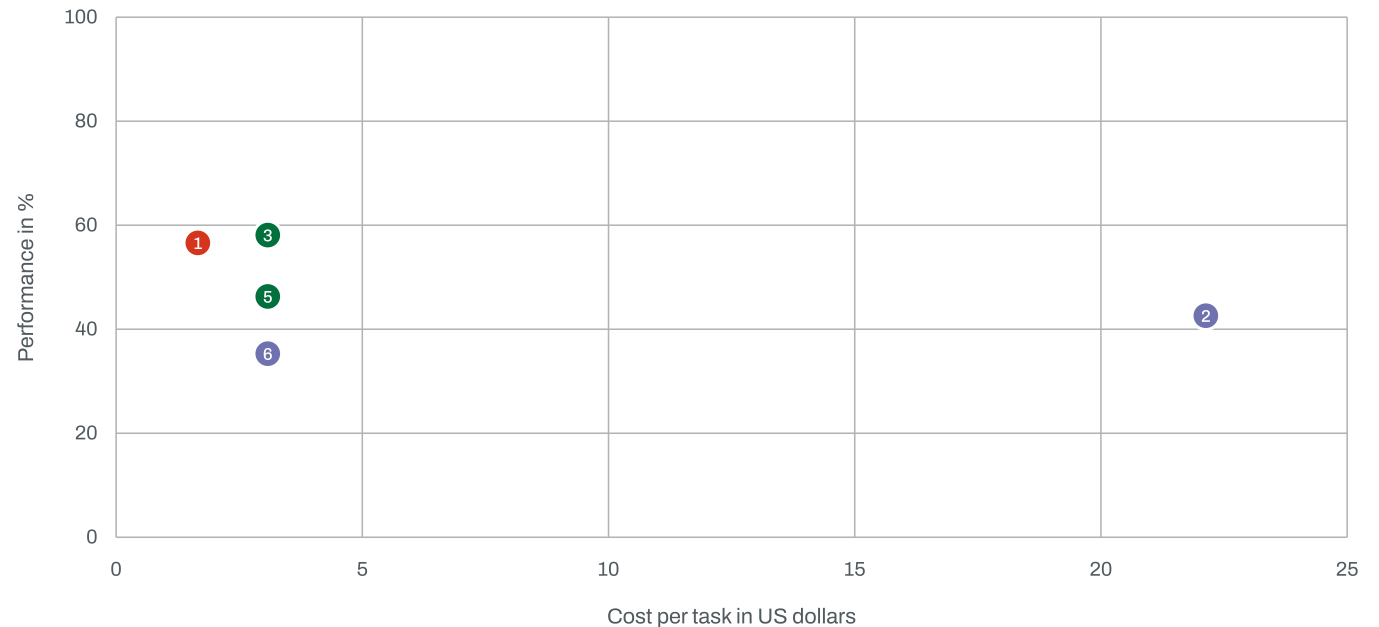
Limit

The site alternates between 128 and 136 samples. Some reported cost differences need more explanation.

XOR results

Dataset or repository

Published results



Point	Model	Harness	Result · cost
1	Claude Opus 4.6	Claude Code	56.6% · \$1.66
2	Claude Opus 4.6	OpenCode	42.6% · \$22.13
3	GPT-5.2	Codex	58.1% · \$3.08
4	GPT-5.2	OpenCode	46.3% · \$3.08
5	GPT-5.2-Codex	Codex	46.3% · \$3.08
6	GPT-5.2-Codex	OpenCode	35.3% · \$3.08

WildClawBench tested 60 long-horizon tasks: changing the harness produced double-digit score shifts and changed the winner by model.

What they did

InternLM evaluated 60 bilingual, multimodal and long-horizon tasks in harness-specific container images. Deterministic checks, environment checks and model judges contributed to the overall score.

Tasks	60
Repeated trials	not clear
Grader	Deterministic checks, environment checks and model judges

What we observe

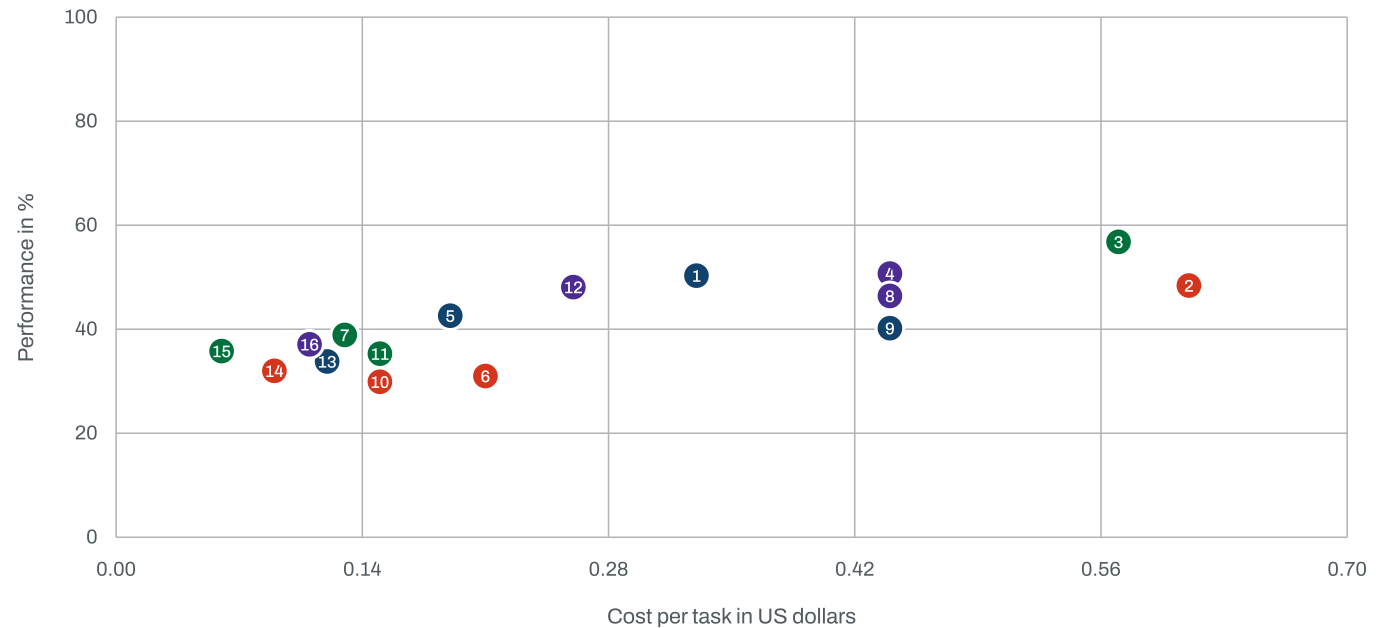
The best harness changed with the model. The source reports double-digit score changes from switching harness.

Limit

Each harness image has different tools and prompt conventions. Repeat counts are not clear in the headline table.

InternLM results
Dataset or repository

Published results



Point	Model	Harness	Result · cost	Point	Model	Harness	Result · cost
1	GPT-5.4	OpenClaw	50.3% · \$0.33	9	MiMo V2 Pro	OpenClaw	40.2% · \$0.44
2	GPT-5.4	Claude Code	48.4% · \$0.61	10	MiMo V2 Pro	Claude Code	29.9% · \$0.15
3	GPT-5.4	Codex	56.8% · \$0.57	11	MiMo V2 Pro	Codex	35.3% · \$0.15
4	GPT-5.4	Hermes Agent	50.7% · \$0.44	12	MiMo V2 Pro	Hermes Agent	48.1% · \$0.26
5	GLM 5	OpenClaw	42.6% · \$0.19	13	MiniMax M2.7	OpenClaw	33.8% · \$0.12
6	GLM 5	Claude Code	31% · \$0.21	14	MiniMax M2.7	Claude Code	32% · \$0.09
7	GLM 5	Codex	38.9% · \$0.13	15	MiniMax M2.7	Codex	35.8% · \$0.06
8	GLM 5	Hermes Agent	46.4% · \$0.44	16	MiniMax M2.7	Hermes Agent	37.1% · \$0.11

CORE-Bench Hard tested 45 scientific reproductions: Claude Code more than doubled verified success while costing less than CORE-Agent.

What they did

Princeton HAL asked agents to reproduce 45 research artefacts. The leaderboard verifies whether each system recreated the required scientific result and reports total cost.

Tasks	45
Repeated trials	no
Grader	Research artifact reproduction

What we observe

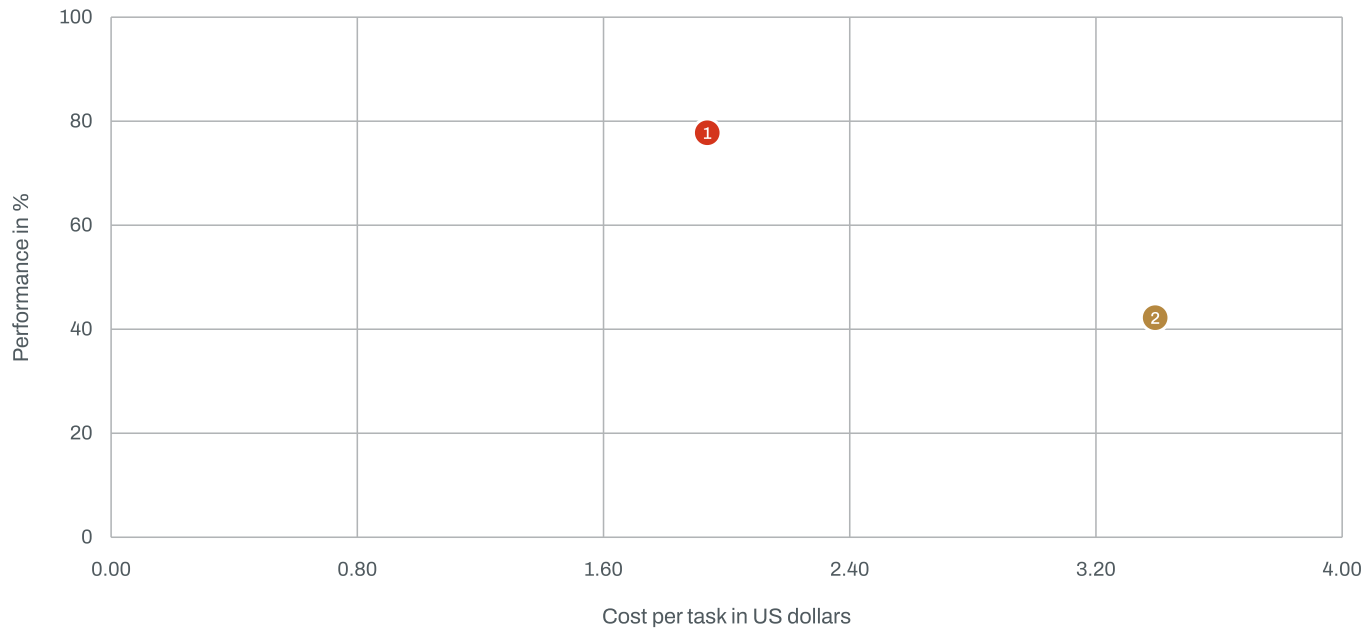
Claude Code more than doubled the verified success of CORE-Agent for Claude Opus 4.5 while costing less.

Limit

There is one run per configuration. The systems differ across retrieval, tools and context policy.

Princeton HAL results
Dataset or repository

Published results



Point	Model	Harness	Result · cost
1	Claude Opus 4.5	Claude Code	77.8% · \$1.94
2	Claude Opus 4.5	CORE-Agent	42.2% · \$3.39

The Scaffold Effect ran 300 trials: Goose used about 41 times fewer reported tokens than OpenCode, with pass rates within 2 to 8 points.

What they did

The study ran 2 models through Goose, OpenCode and OpenHands-SDK on a stratified 50-task slice of Terminal-Bench Pro, with identical environments and a 40-turn budget.

Tasks	50
Repeated trials	no
Grader	Terminal-Bench Pro task checks

What we observe

Resource use separated the harnesses more clearly than pass rate. OpenCode used about 41 times as many reported tokens as Goose.

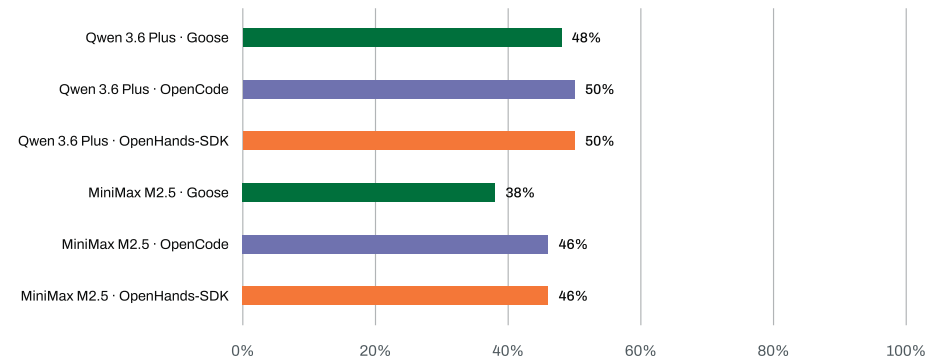
Limit

There is one run per task and configuration. Token accounting differs by harness and is incomplete; cost is missing for 2 harnesses.

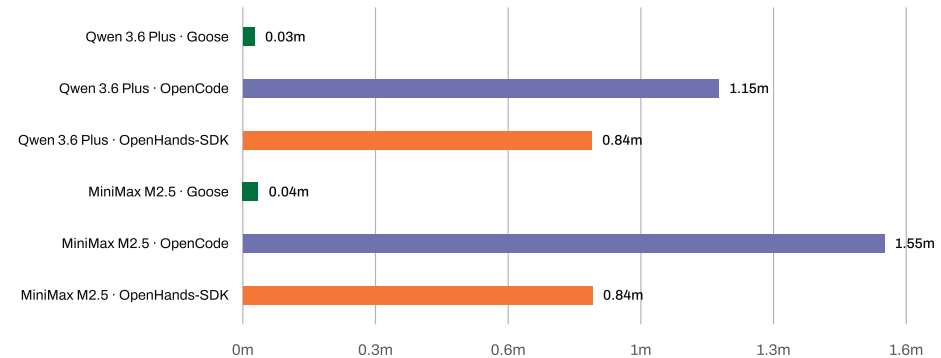
Synapse Social results
Dataset or repository

Published results

Pass rate



Reported tokens per solved task



Both scales start at zero. Token accounting is harness-reported and incomplete, so treat the efficiency contrast as diagnostic.

HAL compared 15 models on 50 SWE-bench tasks: SWE-Agent usually solved more, while HAL Generalist usually cost less.

What they did

Princeton HAL ran the same 50 SWE-bench Verified Mini tasks under SWE-Agent and HAL Generalist for 15 models and effort settings, reporting pass rate and total provider cost.

Tasks	50
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Repeated trials	no
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Grader	SWE-bench repository tests
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What we observe

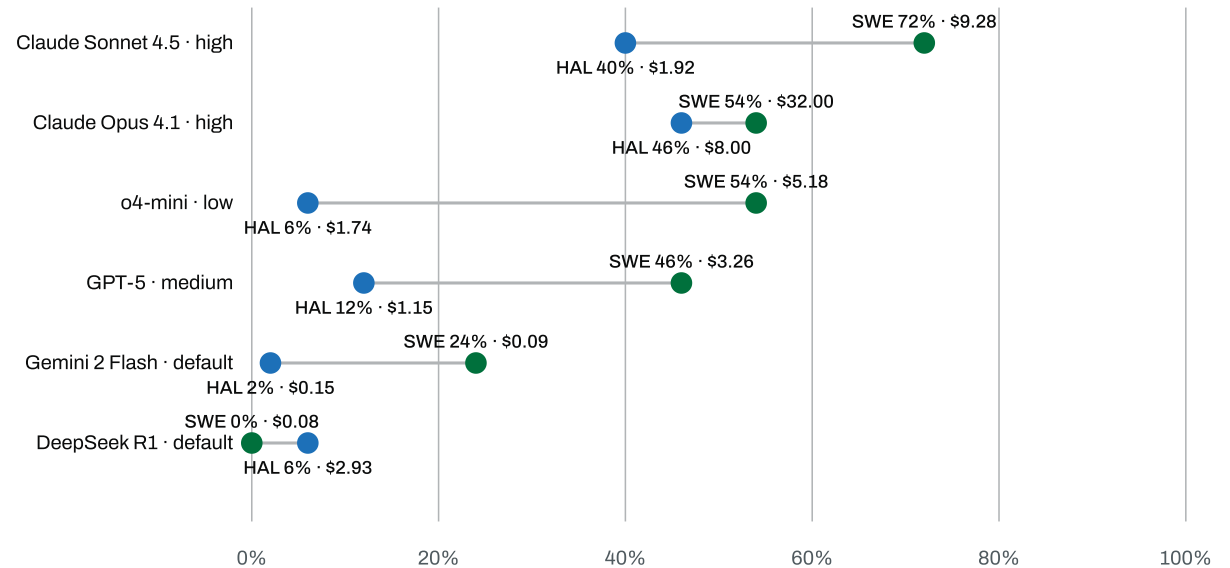
SWE-Agent led 14 of 15 matched pairs on pass rate. HAL Generalist was often cheaper, showing a strong quality-cost trade-off rather than a universal harness win.

Limit

Most entries have one evaluation run. SWE-Agent cost excludes caching, and the archived trace repository has no licence file.

Princeton HAL results
Dataset or repository

Published results



Six representative pairs are shown on a zero-to-100 performance scale; all 15 pairs are in the downloadable observations. Dollar labels are published total cost divided by 50 tasks.

HarnessBench tested 27 recent bug fixes: several configurations finished within one task of the leader, but runtime differed more clearly.

What they did

HarnessBench selected 27 recent bug-fix issues from 9 open-source repositories. Hidden tests checked the core fix and regressions, with one run for each configuration.

Tasks	27
Repeated trials	no
Grader	Hidden core and regression tests

What we observe

Several conditions finished within one task of the observed leader. Runtime differences were clearer than pass-rate differences.

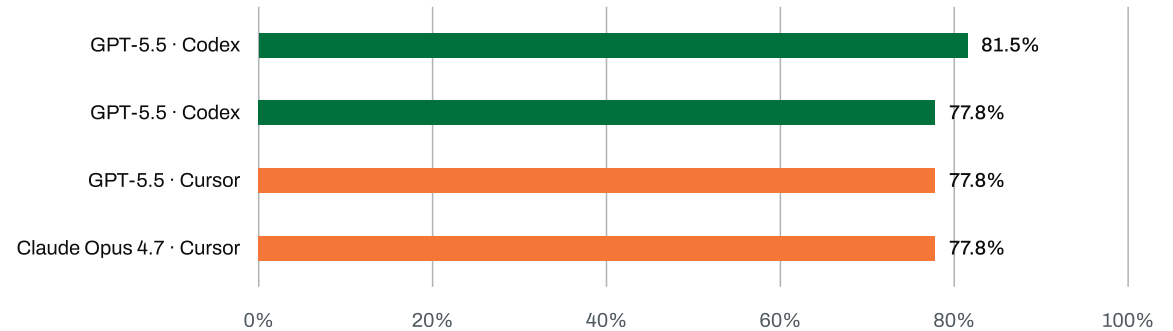
Limit

The author states that the 27-task pass-rate differences are not statistically significant.

Nyosegawa results

Dataset or repository

Published results



A-CODE-CLI repeated 10 web scenarios: OpenCode had a 2.4-point backend lead and cost 44% less than Claude Code.

What they did

AIMultiple ran 10 full-stack web development scenarios 3 times with Claude Sonnet 4.6. Backend checks and Playwright tests measured correctness and user-interface behaviour.

Tasks	10
Repeated trials	yes
Grader	Backend checks and Playwright user-interface checks

What we observe

OpenCode had a small backend lead over Claude Code and cost less in the matched Sonnet 4.6 comparison.

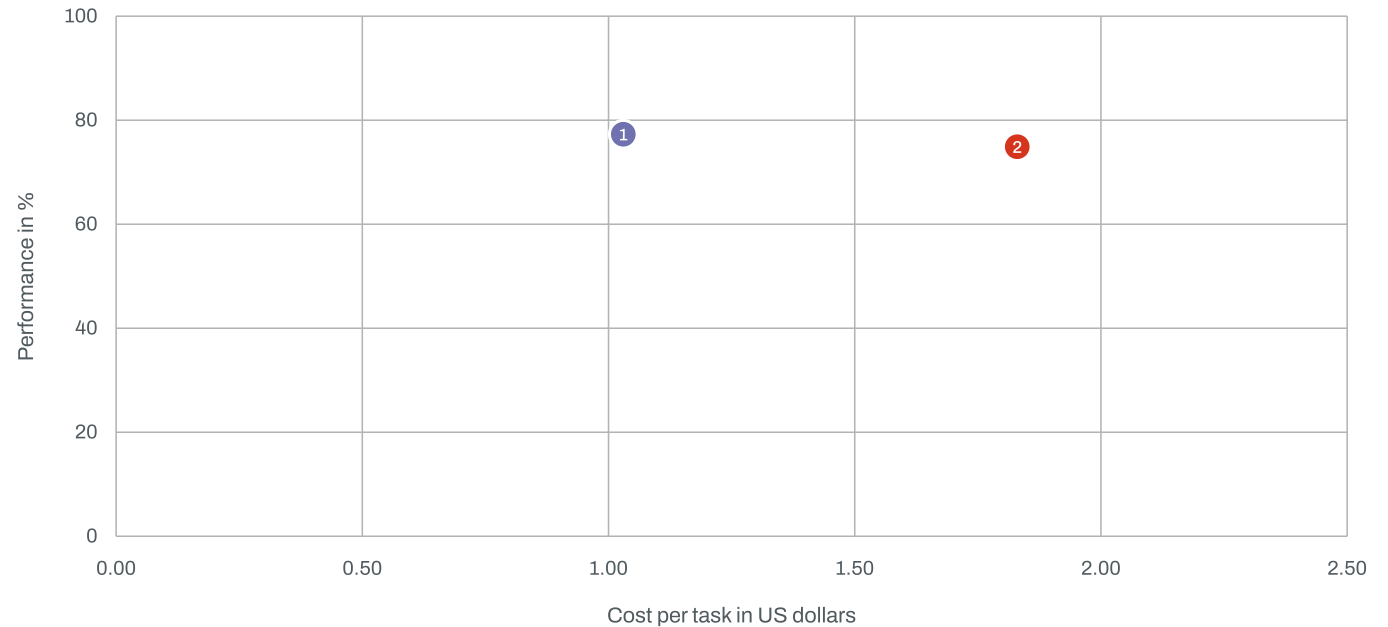
Limit

The benchmark has only 10 scenarios. Only one example task is public.

AIMultiple results

Dataset or repository

Published results



Point	Model	Harness	Result · cost
1	Claude Sonnet 4.6	OpenCode	77.3% · \$1.03
2	Claude Sonnet 4.6	Claude Code	74.9% · \$1.83

OpenBench repeated 3 coding tasks: every harness passed, so the useful difference was efficiency rather than correctness.

What they did

OpenBench ran 3 repository coding tasks 3 times with GPT-5.5 at medium effort. Executable checkers awarded partial credit, while the study also recorded wall time and token use.

Tasks	3
Repeated trials	yes
Grader	Executable checkers with partial-credit support

What we observe

All clean harnesses passed every task. Pi and Cursor used less time and fewer tokens.

Limit

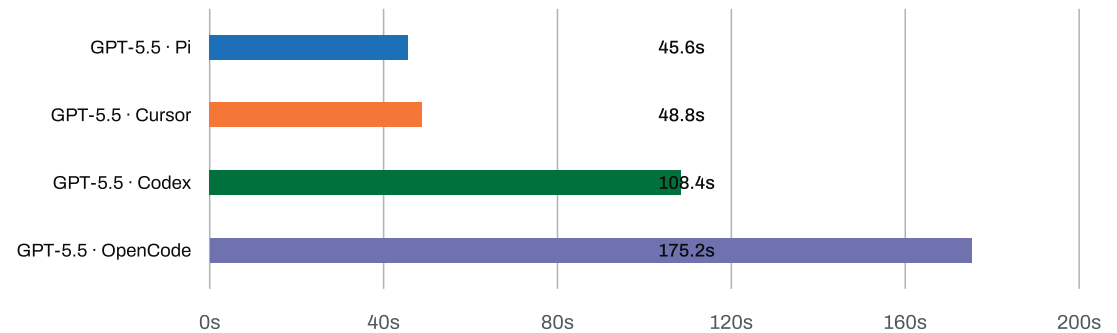
The tasks were too easy to separate frontier harnesses on correctness.

OpenBench results

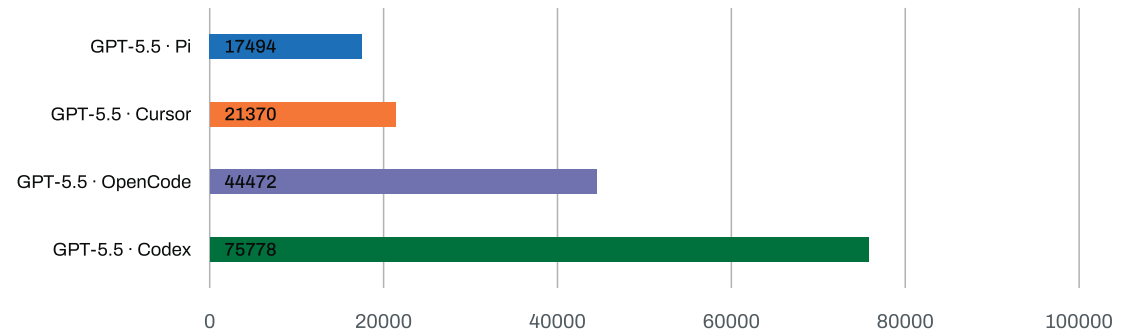
Dataset or repository

Published results

Mean wall time



Tokens per solve



Portkey measured 2 trivial requests: fixed input overhead ranged from about 2,600 tokens for Pi to 27,000 for Claude Code.

What they did

Portkey routed 2 trivial prompts through its gateway and recorded the input payload sent by each harness. The exercise isolates fixed prompt overhead rather than coding performance.

Tasks	2
Repeated trials	no
Grader	Input-token capture

What we observe

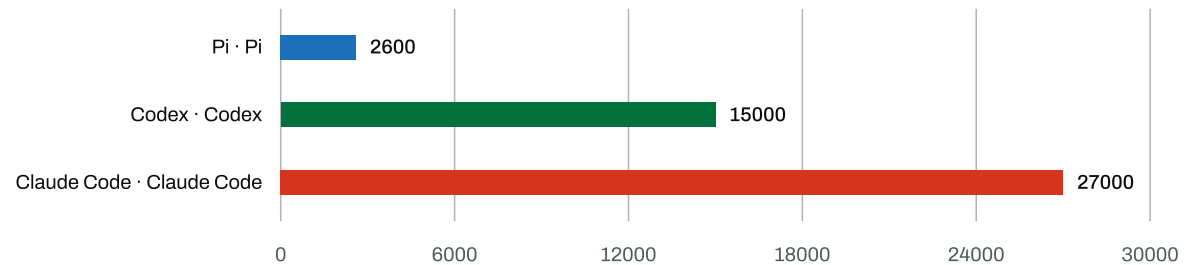
The fixed input payload ranged from about 2,600 tokens for Pi to 27,000 for Claude Code.

Limit

This is an overhead diagnostic. It does not measure coding quality or full-session economics.

Portkey results

Published results



This diagnostic measures fixed input overhead. It does not measure coding quality.

Evidence matrix

23 studies contain at least one matched-model harness comparison. Portkey is an efficiency diagnostic, not a quality benchmark.

Dates are the publication or public-release date of the comparison used here. For live leaderboards, they mark when the referenced comparison first appeared.

No.	Study	Task surface	Model fixed	Includes Pi	Repeats	Cost
01	Databricks coding-agent benchmark	Recent human-written pull requests from a multi-million-line internal codebase	yes	yes	not clear	yes
02	GitHub Copilot harness evaluation	Five coding and terminal benchmark suites, including SWE-bench and Terminal-Bench 2	yes	no	yes	yes
03	Terminal-Bench 2.1	Containerized command-line tasks with deterministic final-state checks	yes	no	yes	yes
04	Coding Agent Index	321 tasks across DeepSWE, Terminal-Bench v2 and SWE-Atlas-QnA	yes	no	yes	yes
05	Agents' Last Exam	Long-horizon professional workflows across CLI, GUI, files, applications and web research	yes	no	not clear	yes
06	DrugDiscoveryBench	Expert-curated drug-discovery and life-sciences workflows	yes	yes	yes	yes
07	Claw-SWE-Bench	Multilingual GitHub issue-resolution tasks	yes	no	no	yes
08	Harness-Bench paper	Executable office, retrieval, analytics, SRE and software-engineering workflows	yes	no	no	no
09	WANDR system comparison and delivery ablation	Wide-and-deep web research requiring large, evidence-backed collections	yes	no	no	yes
10	Warden benchmark	Known-vulnerability recognition in files tied to historical Sentry vulnerabilities	yes	yes	no	yes
11	Agent Security League	Real vulnerability-fixing tasks with hidden functional and security tests	yes	no	no	no
12	Harvey LAB harness ablation	Held-out legal workflows scored against expert-written criteria	yes	no	no	no
13	Harness optimization on LAB	Realistic legal matters requiring evidence-backed deliverables	yes	yes	no	yes
14	LatchBio behavioural-prior study	Short-horizon preclinical pharmacology and long biological analysis workflows	yes	yes	yes	yes
15	CSI scaffold comparison	CyBench capture-the-flag security challenges	yes	no	no	yes
16	Vulnerability-Agent-Bench	Real vulnerability repairs in isolated containers	yes	no	no	yes
17	WildClawBench	Bilingual, multimodal and long-horizon agent tasks	yes	no	not clear	yes
18	CORE-Bench Hard	Scientific reproducibility tasks	yes	no	no	yes

No.	Study	Task surface	Model fixed	Includes Pi	Repeats	Cost
19	The Scaffold Effect	A 50-task slice of Terminal-Bench Pro	yes	no	no	no
20	HAL SWE-bench Verified Mini	A fixed 50-task subset of SWE-bench Verified	yes	no	no	yes
21	HarnessBench	Recent bug-fix issues from 9 open-source repositories	yes	no	no	yes
22	A-CODE-CLI	Ten full-stack web development scenarios	yes	no	yes	yes
23	OpenBench M4.5	Three repository coding tasks with executable checkers	yes	yes	yes	no
24	Harness Tax	Two trivial prompts routed through a gateway	no	yes	no	no

[How to read harness comparisons](#) · [Download observations](#) · [Download study records](#)

Screened but not counted

These sources are relevant to the search, but they do not currently isolate a harness effect or they duplicate an included benchmark family.

Source	Date	Disposition	Reason
Factory Terminal-Bench article	25 Sept 2025	Not counted separately	The headline Droid comparisons use different model variants from Claude Code and Codex. Valid Terminal-Bench comparisons are covered under the benchmark family.
Harvey Legal Agent Benchmark introduction	6 May	Context only	The launch defines the legal benchmark but publishes no comparison that changes the harness while holding the model fixed.
Harvey LAB initial results	26 May	Excluded	It compares models under a fixed harness. The later 22 June harness ablation is included instead.
FrontierSWE	Apr 2026	Watchlist	The benchmark repeats 17 long-horizon tasks, but each model currently appears under only one harness, so the harness effect cannot be isolated.
Artificial Analysis general agents catalogue	Not dated	Excluded	This is a product catalogue, not a benchmark. Alternate Coding Agent Index URLs are merged into the single included study.
SWE-bench Pro harness-model pairing comparison	7 Aug	Watchlist	The 10-harness, 2-model comparison isolates harness effects across 250 tasks, but the post links no raw table or repository and publishes only selected exact values. Add observation rows when the underlying data is public.

Data and source access

External datasets remain under their publishers' terms. This repository stores derived observations and links to the original data.

Source	Access	Licence note	Links
Benchmarking coding agents on Databricks' multi-million-line codebase	No public task-level dataset found	Not stated for benchmark data	Results
Warden benchmark	Public aggregate table	Not stated for benchmark results	Results · Dataset or repository
LatchBio behavioural-prior study	Public TxBench-PP and scBench-Long JSON summaries and trajectories; aggregate prompt-ablation table in the article	Check the linked benchmark repositories and dataset terms	Results · Dataset 1 · Dataset 2
Harness optimization on the Legal Agent Benchmark	Public article source, harness code, frozen split, dev and test results, and proposer traces	Article text and diagrams are CC BY 4.0; check the linked code and benchmark terms	Results · Dataset or repository
OpenBench	Public repository with committed data for several milestones	MIT	Results · Dataset or repository
DrugDiscoveryBench	Tasks and code are public; rubrics and reference answers are gated on Hugging Face	MIT for repository code and tasks; separate terms apply to bundled data and gated material	Results · Dataset or repository
Artificial Analysis Coding Agent Index	Live public result page and methodology; no stable bulk dataset found	Result-data licence not stated; site terms apply	Results
Endor Labs Agent Security League	Public leaderboard and articles	Not stated for benchmark data	Results
WANDR benchmark and output-delivery ablation	Public paper, 500 task packages, evaluation code and aggregate tables; evaluated run artefacts are not included	Apache 2.0 repository and CC BY 4.0 paper; third-party task sources retain their own terms	Results · Dataset or repository
Vulnerability-Agent-Bench	Public explorer and linked Weights and Biases project	Not stated for benchmark data	Results · Dataset or repository
WildClawBench	Public code, tasks, runtime images and dataset	Check the repository and dataset cards	Results · Dataset or repository
HarnessBench	Public code and results	Check the repository licence	Results · Dataset or repository
CORE-Bench Hard	Public harness, leaderboard and encrypted traces	Check the repository licence	Results · Dataset or repository
A-CODE-CLI	Public report and one example task	Check the example repository licence	Results · Dataset or repository
The harness tax	Public article with aggregate values	Not stated for request logs	Results

Source	Access	Licence note	Links
GitHub Copilot agentic harness evaluation	Public methods and aggregate charts; no raw result rows found	Not stated for benchmark result data	Results
Terminal-Bench 2.1	Public leaderboard, paper, experiments and submissions	Apache-2.0 repositories; paper CC BY 4.0	Results · Dataset or repository
Agents' Last Exam	Public paper, HTML tables, code and data	Data and paper CC BY 4.0; software Apache-2.0	Results · Dataset or repository
Claw-SWE-Bench	Public paper, adapters, dataset and result artefacts	MIT repository; check dataset card for task data terms	Results · Dataset or repository
Harness-Bench: Measuring Harness Effects on LLM Agents	Public paper, tasks, code and aggregated JSON	Paper CC BY 4.0; repository has no licence file as captured	Results · Dataset or repository
Harvey LAB harness ablation with Applied Compute	Public aggregate chart; baseline benchmark and harness are public, ablation split is not	MIT for public LAB repository; ablation result-data licence not stated	Results · Dataset or repository
Cybersecurity scaffold comparison	Public paper and project page; no dedicated experiment artefact found	Paper CC0; related code has mixed MIT and research-only terms	Results
The Scaffold Effect	Public raw results, prompts and analysis	MIT repository	Results · Dataset or repository
HAL SWE-bench Verified Mini	Public leaderboard and downloadable traces	Archived trace repository has no licence file as captured	Results · Dataset or repository